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/ **Practice Exam - 1**

Correct Answer

Next Q

Practice Exam - 1

Question 29 of 117



A 79-year-old man with permanent atrial fibrillation, hypertension, nonischemic cardiomyopathy with a left ventricular ejection fraction partially recovered to 40%, and biventricular implantable cardioverter- defibrillator presents to your office with progressive shortness of breath over the last month while ascending one flight of stairs. Prior to this he was well compensated. Medications include carvedilol 6.25 mg twice daily, sacubitril/valsartan 49/51 mg twice daily, spironolactone 25 mg daily, dapagliflozin 10 mg daily, torsemide 20 to 40 mg daily, and apixaban 5 mg twice daily. Recent coronary angiography demonstrates no obstructive coronary disease. He has been unable to tolerate further uptitration of medical therapy because of hypotension.

On examination, his blood pressure is 95/74 mmHg, heart rate is 72 beats per minute, respiratory rate is 16 respirations per minute, and oxygen saturation is 98% on room air. He exhibits no jugular venous distension. Cardiac examination demonstrates a laterally displaced point of maximum impulse, and a III/VI holosystolic murmur best heard at the apex. There is no peripheral edema. The remainder of the examination is benign.

Transesophageal echocardiography demonstrates a mitral regurgitant orifices area of 72 mm², a regurgitant volume of 65 ml, and a vena contracta jet width of 9 mm.

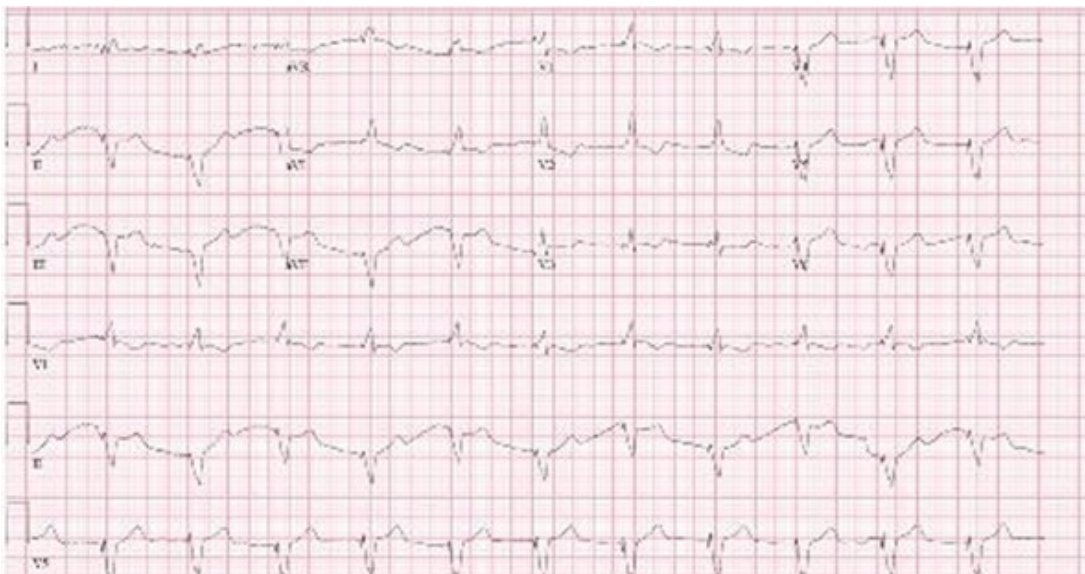


Figure 1. Electrocardiogram



Figure 2. Transthoracic Echocardiogram

The next best step in the plan for caring for this patient is to:

- A. Pursue echocardiographic dyssynchrony study
- ☒ B. Implantation of a pulmonary artery pressure monitor
- C. Implantation of a left atrial appendage occlusion device
- ☒ D. Initiate transcatheter edge-to-edge repair (TEER) of the mitral valve evaluation
- E. Initiate ventricular assist device evaluation

Commentary

References

Testing Point: The Board-certified AHFTC specialist should know that transcatheter edge to edge repair (TEER) of the mitral valve is an indicated procedure for severe secondary mitral regurgitation.

Answer: D. Initiate transcatheter edge-to-edge repair (TEER) of the mitral valve evaluation.

This patient has severe secondary mitral regurgitation. The transthoracic echocardiogram clips demonstrate a dilated left ventricle, and tenting of the mitral valve, indicating restricted leaflet motion due to tethering of the valve leaflets. This is considered secondary (or functional) mitral regurgitation. The COAPT trial using the MitraClip device to effect transcatheter edge-to-edge repair (TEER) of the mitral valve demonstrated that patients with moderate-to-severe (3+) or severe (4+) mitral regurgitation treated with MitraClip in addition to optimal medical therapy had reduced heart failure hospitalization or mortality compared to patients on optimal medical therapy alone. Based on this study, the 2020 ACC/AHA guideline for the Management of Valvular Heart Disease recommends that “patients with chronic severe secondary MR related to LV systolic dysfunction (LVEF < 50%) who have persistent symptoms (NYHA class II, III, or IV) while on optimal GDMT for HF (Stage D), TEER is reasonable in patients with...LVEF between 20% and 50%, LVESD ≤ 70 mm, and pulmonary artery systolic pressure ≤ 70 mmHg.”

Answer A: While the QRS is paced and wide, attempts to improve cardiac synchrony after CRT-P implantation using echocardiographic dyssynchrony studies have demonstrated only mixed results.

Answer B: The patient is symptomatic with a euvolemic examination. While implantation of a pulmonary artery pressure monitor may allow tighter management of diuretics, in the setting of untreated severe MR, it is unlikely to lead to significant clinical improvement.

Answer C: There is no evidence for clinical bleeding to suggest that left atrial appendage occlusion would represent an improvement over the current anticoagulation strategy.

Answer E: The patient's age and the presence of an alternative treatment make VAD evaluation less attractive.

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